

Experiment Matrix

1. Purpose

This matrix organizes future experiments by research question, input type, and success metric.

The current comparison snapshot used in the project ranks the screening models as:

1. multimodal_attention

1. transformer

1. vit

1. cnn

1. lstm

2. Multimodal Screening Experiments

| Experiment | Inputs | Model | Metric | Why It Matters |

| CNN baseline | handwriting + audio + errors | `InitialCNNModel` | accuracy / F1 |

Smallest multimodal baseline |

| Text baseline | text + behavior + errors | `InitialLSTMModel` | accuracy / F1 |

Checks text-driven signal strength |

| Full baseline | image + audio + text + behavior + errors | `InitialCNNLSTMModel` |

accuracy / F1 | Early full fusion reference |

| Default multimodal | all modalities | `MultimodalDyslexiaModel` | accuracy / F1 |

Main production-style baseline |

| Transformer fusion | all modalities | `TransformerMultimodalModel` | accuracy / F1 |

Tests stronger text modeling |

| Attention fusion | all modalities | `AttentionMultimodalModel` | accuracy / F1 +

attention analysis | Tests interpretability |

| ViT fusion | all modalities | `ViTMultimodalModel` | accuracy / F1 | Tests

patch-based handwriting modeling |

3. Low-Resource Transfer Experiments

| Experiment | Inputs | Method | Metric | Why It Matters |

| Scratch vs transfer | Bengali manifest | cross-lingual warm start | delta F1 |

Measures transfer benefit |

| Freeze vs finetune | Bengali manifest + source checkpoint | prefix freezing | delta

F1 | Finds stable warm-start recipe |

| Distillation test | Bengali manifest + teacher checkpoint | shared feature distillation

| delta F1 / stability | Tests teacher-guided adaptation |

| Foundation adaptation | Bengali manifest | foundation + adapter | accuracy / F1 |

Tests reusable latent space |

4. Explainability Experiments

| Experiment | Inputs | Method | Metric | Why It Matters |

| Grad-CAM check | handwriting images | `GradCAM` | human plausibility review | Tests

image explanation quality |

| Attention check | multimodal outputs | modality attention | agreement score | Tests

if weights are sensible |

| Text attention check | text sequences | transformer token scores | qualitative review

| Tests token-level interpretability |

| Educational summary review | prediction output | explanation text | teacher usability

feedback | Tests whether explanations are readable |

5. Biomarker Experiments

| Experiment | Inputs | Method | Metric | Why It Matters |

| Biomarker ranking | numeric manifest | `discover\_digital\_biomarkers` | ranking

stability | Finds strongest signals |

| Family comparison | feature families | group analysis | effect size | Compares

handwriting vs speech vs reading |

| Cross-language comparison | Bengali + English | split-wise analysis | rank overlap |

Tests portability of markers |

6. Intervention Experiments

| Experiment | Inputs | Method | Metric | Why It Matters |

| Plan selection | learner profile | `InterventionPolicy` | chosen action consistency

| Tests policy behavior |

| Reward update | session progress | policy update loop | reward improvement | Tests

learning signal |

| Tutor adaptation | tutor state | `AdaptiveTutorAgent` | session score gain | Tests

adaptive practice value |

7. Suggested Reporting Metrics

- accuracy

- F1 score

- recall for elevated-risk cases

- MAE for regression tasks

- explanation usefulness

- modality-attention stability

- biomarker rank stability
- intervention improvement across sessions
- cross-validation selection accuracy
- hard holdout F1 / precision / recall